DSA06 DATA HANDLING AND VISUALIZATION | Activity: Identify Seasonality Patterns | Slot C

R.No: 25 Reg. No: 192473018 Name: N LAHARI

Time: 20 to 25 minutes Total: 20 marks Tool: R (ggplot2 / base R). Work alone; every student's series is different. Write your answers and paste your plots and code.

Series SP-C25: Bus ridership [20 marks]

Scenario: You are the analyst for a city bus depot. The series riders (Riders per day) is recorded daily. Management believes it follows a repeating pattern and wants it identified before planning. The cycle length is 7 observations. Run the code below to create the series x and answer the tasks.

Evidence: run this code in R (do not edit it).

```
set.seed(19339)
m <- 7                    # observations per cycle (daily data)
n <- m * 10               # 10 full cycles
t <- 1:n
trend <- 5115.1 + 76.348 * t
riders <- round(trend * (1 + 0.20 * cos(2 * pi * (t - 7) / m)) + rnorm(n, 0, 274.6), 0)
x <- ts(riders, frequency = m, start = c(1, 1))
```

Tasks:

1. Look at the series. Plot x (plot(x) or ggplot2). State n, the cycle length, the number of full cycles, and whether the overall trend is rising, falling or flat (fit lm(riders ~ 1:n) and report the slope per day). [3]
2. The seasonal pattern. Use decompose(x) and extract $figure (the 7 seasonal values). Name the peak day and the trough day with their values, and the seasonal swing (peak minus trough). [4]
3. Is it consistent? Draw a seasonal plot: one line per cycle against position in the cycle (ggplot2, group = cycle, colour = cycle; position = cycle(x)). In how many of the 10 cycles is the highest value in the peak day from Task 2? Is the pattern consistent? [3]
4. Confirm with autocorrelation. Detrend (d <- resid(lm(riders ~ 1:n))) and compute acf(d, lag.max = 7, plot = FALSE) using observation lags. Report the autocorrelation at lag 7 and at lag 3, and compare with the significance bound 2/sqrt(n). What do the two values say about seasonality? [4]
5. Additive or multiplicative? Compute the range (max minus min) of riders in the first cycle and in the last cycle and their ratio (last / first); decide the type and justify [2]. Remove the seasonal part with decompose(x, type = ...) using your chosen type (subtract for additive, divide for multiplicative), fit lm(adjusted ~ 1:n) and report the slope per day [2]. Write one sentence of advice to management based on your findings [2]. [6]

Remember: *seasonality repeats at a fixed, known period; a trend is not a season; if the swings grow with the level, the season is multiplicative.*

Code of Conduct: I certify that this work is my own. Signature: _____

22/10/26

Activity - 08

N. Lahari

Identify Seasonality Patterns

1) Time Series and trend

* No of observations, n = 70
* Cycle length = 7 observations
* No of full cycles = 70/7 = 10
* Overall trend : Rising trend
* The underlying trend equation is 5115.1 + 76.348t, so, the expected trend increases by approximately 76.348

2) Seasonal pattern

The series contains a repeating seasonal pattern with a period of 7 days. The seasonal component can be extracted using decompose(x).

* Peak day : The day with the highest seasonal value.
* Trough day : the day with the lowest seasonal value.
* Seasonal swing = Peak seasonal value - Trough seasonal value

3) Seasonal plot and consistency

=> A seasonal plot compares the 10 cycles against the seven positions in each cycle. The lines should show a broadly repeating weekly pattern, although random noise causes differences between cycles.

=> To answer the second part, count how many have their highest observed ridership at the peak position identified in Question

4) Confirm with autocorrelation

Autocorrelation is used to check whether the ridership pattern repeats after a fixed no of observations.

=> The series is detrended by fitting a linear model & calculating the residuals. The autocorrelation is then calculated

at lags 7 & 3.

The significance bound is: $\frac{2}{\sqrt{n}} = \frac{2}{\sqrt{70}} \approx 0.239$

* If the autocorrelation at lag 7 is greater than 0.239, it supports a repeating weekly pattern.
* A Strong autocorrelation at lag 7 supports the presence of seasonality.

5) Additive or multiplicative seasonality

The series is generated using a multiplicative seasonal pattern because the trend is multiplied by a seasonal factor.

The range for each cycle is calculated as:

$$\text{Range} = \text{Maximum riders} - \text{Minimum riders}.$$

The ratio is:

$$\text{Range ratio} = \frac{\text{Last cycle range}}{\text{First cycle range}}$$

Advice to Management: The bus depot should consider the repeating seven-day ridership pattern and the rising overall trend when planning bus schedule and allocating vehicles.

R-Code

```
s <- decompose(x)$figure
Print(s)
Cat("Peak:", which.max(s), max(s))
Cat("\n Trough:", which.min(s), min(s))
Cat("\n Swing:", max(s) - min(s))
d <- resid(lm(riders ~ t))
a <- acf(d, lag.max = 7, plot = FALSE)$acf
Cat("\n ACF lag 7:", a[8], "\n ACF lag 3:", a[4])
Cat("Bound:", 2/sqrt(n))
r <- sapply(1:10, function(i) diff(range(riders[((i-1)*7+1):(i*7)])))
Cat("\n First range:", r[1], "Last range:", r[10])
adj <- riders / rep(s, 10)
cat("\n Slope:", coef(lm(adj ~ t))[2])
```

Bus Ridership Time Series.

